

Title: Q-Mutate: A Computational Analysis of Quantum Proton Tunneling in DNA

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This paper presents a quantum mechanical simulation of DNA base-pair stability. By solving the Time-Independent Schrödinger Equation, I calculated the probability of proton transfer in hydrogen bonds, identifying a significant quantum-mechanical pathway for spontaneous genetic mutations at biological temperatures (310 K).

I utilized the Finite Difference Method to discretize the Hamiltonian operator on a spatial grid of 2000 points. The potential energy was modeled as a Quartic Double-Well, representing the chemical environment of a Guanine-Cytosine (G-C) base pair.

The system is governed by the equation:

$\hat{H}\psi = E\psi$ Where $\hat{H}$ is the Hamiltonian matrix containing kinetic and potential energy terms. I employed a Sparse Matrix Eigenvalue Solver (`scipy.sparse.linalg.eigsh`) to determine the ground state and first excited state energies.

This project confirms that Quantum Tunneling is a viable mechanism for spontaneous point mutations. While classical biology treats DNA as a static database, this research highlights its nature as a dynamic quantum system. Understanding these "Quantum Hotspots" is critical for the future of:

Precision Oncology: Identifying where mutations are most likely to occur.

Quantum Bioinformatics: Developing new algorithms to predict genetic decay.

Future Work:

I intend to expand this model to include Environment-Induced Decoherence, analyzing how the surrounding cellular water molecules affect the tunneling rate.

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Q-MUTATE CORE: CONVERGED METRICS

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Barrier Peak Height (V_{max}): 0.142058 eV
Ground State Energy (E_0): 0.065263 eV
1st Excited Energy (E_1): 0.069176 eV
Energy Splitting (ΔE): 0.003913 eV

Tunneling Frequency: 9.461e+11 Hz
Tunneling Frequency: 0.946 THz

Optimized Parameter a: 2.887676 eV/Å⁴
Optimized Parameter b: 0.470955 Å

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